

Windmills at Sea

Project for modelling 4 (TW2-41)

1. Introduction

There is an increasing activity taking place in our (coastal) seas, examples are the construction of pipelines to transport oil and gas, and the development of **windmill farms**. Windmills become bigger and bigger (see Fig. 1) and windmill farms are more and more developed in **locations** with extreme wave conditions. Of course, it is important that these structures can cope with these conditions. Examples of testing the stability and robustness of windmills can be seen in the movies and information leaflets on **the website of MARIN**.

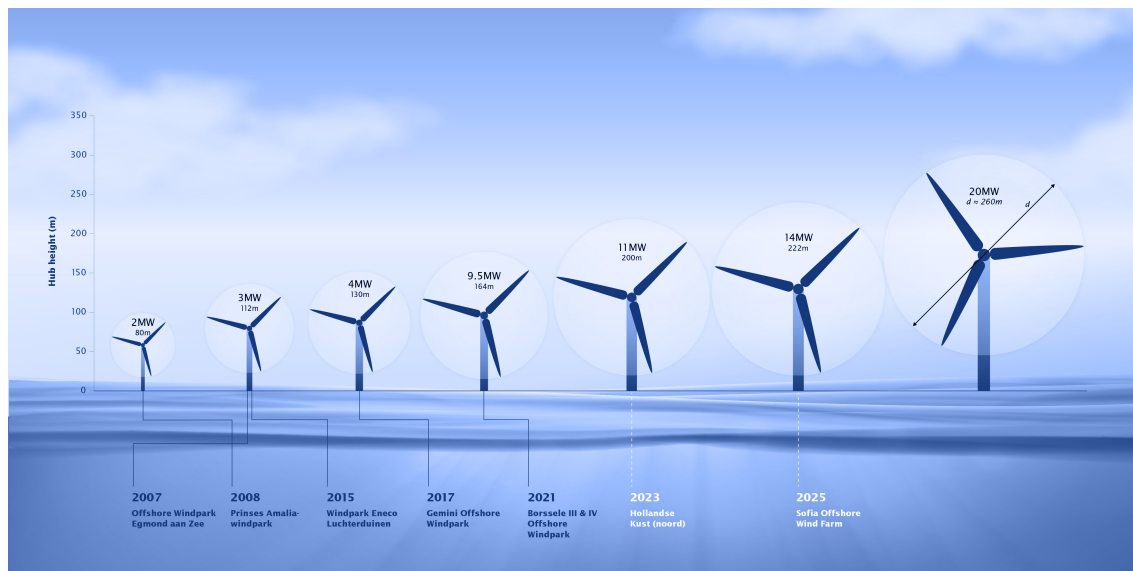


Figure 1. Increase in size of the windmills over the years.

With these challenges in mind, the following research question can be identified:

What is the influence of different forcings and boundary conditions on a windmill, modelled as a vertical cantilever beam?

2. Modelling

To get insight in the stability of the windmills, we will develop a mathematical model that is fit to answer the research question you have formulated. To develop such a model, we will use the 'modelling cycle'. As a first step, the windmill will be approximated as a cantilever beam, see Fig. 2. The equation describing the spatio-temporal evolution of the displacement of the beam is captured by a partial differential equation, second order in time and fourth order in space. The beam is forced by the waves slamming into the beam. The force, exerted by the waves on the beam is obtained by modeling the wave motion using

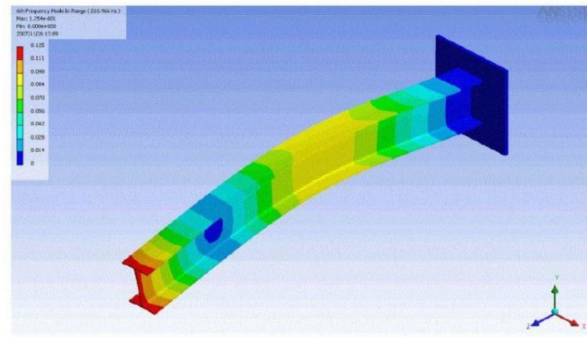


Figure 2. Cantilever beam, attached to a wall.

potential theory. This results in a forced partial differential equation that can be studied using analytical techniques similar to those studied in the Partial Differential Equations course (TW2-23), Q2. Next, you will make specific extensions necessary to answer your explicit research question. Depending on the research questions you pose, numerical extensions can be considered, but this is not necessary.

3. Project

In this project you develop a model to study the stability of a cantilever beam. We take several steps in this:

1. Derive the equation for the cantilever beam and the appropriate boundary conditions, getting insight in the approximations you (have to) make. Given initial conditions.
2. Develop a model for the wave motion and the wave force exerted by the water motion on the beam.
3. With the insight gained, formulate the research question you want to solve.
4. A solution of the (forced) beam equation will be constructed using semi-analytical methods.
5. Based on your research question, you choose how to extend the basic cantilever model. Following the modelling cycle, you are encouraged to take one step at a time. You investigate which mathematical methods you may use to analyse your new model, construct, and verify your solutions.
6. You use your model to answer (part of) the research question and write down your findings in a report.

4. Time Schedule

The table below indicates the approximate time schedule for this project

Week	Activity	Deliverable/feedback
Week 1	Develop model components: Beam model: Euler-Bernouilli beam equation, also see for example Rao (2018) What do the parameters mean? Water motion: Linear wave theory, see also for example Holthuijzen (2007) Wave forcing: Morison Equation	
Week 2	Develop model and get analytical solutions. Continue with the tasks of Week 1. Solve the Euler-Bernouilli beam equation. (use techniques from TW2070 (PDE)) Solve for the water motion.	
Week 3	Continue solving the equations, including wave forcing term. Start writing.	Hand in individual assignment.
Week 4	Proposal for research question and model extension Continue writing, including first version of introduction.	Discuss model extension with advisor.
Week 5	Start working on model extension	Hand-in current document for feedback.
Week 6	Visit to MARIN , Continue analysing model extension	Receive feedback. Small presentation at Marin
Week 7	Continue analysing model extension Write report.	Receive feedback
Week 8	Give presentation. Write report	
Week 9/10	Write report	
Week 10/11	Write report.	Hand in Report

5. Contact

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6. Reference

Rao, S. *Mechanical Vibrations*, 2018, Addison–Wesley.
Holthuijsen, L.H., *Waves in Oceanic and Coastal Waters*, 2007, Cambridge University Press.